A Concise Derivation of Classical Multidimensional Scaling

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Goal

Given n objects with pairwise *squared* dissimilarities

$$d_{ij}^2 \in \mathbb{R}_{\geq 0}, \qquad i, j = 1, \dots, n,$$

Classical MDS (cMDS) aims to find low-dimensional coordinate vectors

$$z_i \in \mathbb{R}^k, \qquad i = 1, \dots, n, \quad k \ll n,$$

such that the Euclidean distances in the embedding space recover the given dissimilarities as faithfully as possible:

$$\|z_i - z_j\|^2 \approx d_{ij}^2.$$

The key insight of cMDS is to work with *inner products* rather than distances directly, because inner products admit a clean matrix factorisation. The derivation therefore proceeds in two phases: (1) convert d_{ij}^2 into a matrix of inner products, and (2) recover the coordinates via eigendecomposition.

Step 1: From Distances to Inner Products

Suppose the (unknown) coordinates $z_1, \dots, z_n \in \mathbb{R}^k$ are *centred*:

$$\sum_{i=1}^{n} z_i = \mathbf{0}. \tag{C}$$

Define the **Gram matrix** $B \in \mathbb{R}^{n \times n}$ by

$$B_{ij} = z_i^\top z_j.$$

Our goal is to express B_{ij} purely in terms of d_{ij}^2, so that we can compute B from the given dissimilarities without knowing z_i.
Expand the squared distance:

$$d_{ij}^2 = \|z_i - z_j\|^2 = \|z_i\|^2 - 2\, z_i^\top z_j + \|z_j\|^2 = B_{ii} - 2B_{ij} + B_{jj}. \tag{1}$$

Sum (1) over $j = 1, \dots, n$ and use the centring condition (C):

$$\sum_{j=1}^{n} d_{ij}^2 = n\, B_{ii} - 2 \sum_j B_{ij} + \sum_j B_{jj} = n\, B_{ii} + \operatorname{tr}(B),$$

because $\sum_j B_{ij} = z_i^\top \sum_j z_j = 0$ by (C). Hence

$$\frac{1}{n} \sum_{j=1}^{n} d_{ij}^2 = B_{ii} + \frac{1}{n} \operatorname{tr}(B). \tag{2}$$

Similarly, summing (1) over i:

$$\frac{1}{n}\sum_{i=1}^{n} d_{ij}^2 = B_{jj} + \frac{1}{n}\,\mathrm{tr}(B). \tag{3}$$

Summing (1) over both i and j:

$$\frac{1}{n^2}\sum_{i,j} d_{ij}^2 = \frac{2}{n}\,\mathrm{tr}(B). \tag{4}$$

Now solve for B_{ij}. From (1):

$$B_{ij} = \tfrac{1}{2}\left(B_{ii} + B_{jj} - d_{ij}^2\right).$$

Substitute (2), (3), and (4) to eliminate B_{ii}, B_{jj}, and $\mathrm{tr}(B)$:

$$B_{ij} = -\tfrac{1}{2}\left(d_{ij}^2 - \frac{1}{n}\sum_{j'} d_{ij'}^2 - \frac{1}{n}\sum_{i'} d_{i'j}^2 + \frac{1}{n^2}\sum_{i',j'} d_{i'j'}^2\right). \tag{5}$$

Why this works: equation (5) eliminates the unknown norms B_{ii}, B_{jj} and the trace by taking row, column, and grand means of d_{ij}^2—a standard *double-centring* operation.

Step 2: Matrix Form of Double Centring

Let $D \in \mathbb{R}^{n\times n}$ denote the matrix of squared dissimilarities, $D_{ij} = d_{ij}^2$, and let

$$H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$$

be the **centring matrix**, where $\mathbf{1} = (1,\dots,1)^\top \in \mathbb{R}^n$. H is symmetric and idempotent ($H^2 = H$); it projects out the mean of any vector it acts on.
Applying the double-centring formula (5) in matrix form (see Appendix A):

$$\boxed{B = -\tfrac{1}{2}\,HDH.} \tag{6}$$

This single expression computes B directly from the observed dissimilarity matrix D.

Step 3: Eigendecomposition of B

Because $B = ZZ^\top$ where $Z = [z_1 \mid \cdots \mid z_n]^\top \in \mathbb{R}^{n\times k}$, the matrix B is *symmetric positive semidefinite* (PSD; see Appendix B).
Applying the eigendecomposition (see Appendix C) to B:

$$B = V\Lambda V^\top,$$

where

$$\Lambda = \mathrm{diag}(\lambda_1,\dots,\lambda_n), \quad \lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_n \ge 0,$$

and $V = [v_1 \mid \cdots \mid v_n]$ is orthogonal ($V^\top V = I$).
To obtain a k-dimensional embedding, retain only the k largest eigenvalues. Partition:

$$\Lambda_k = \mathrm{diag}(\lambda_1,\dots,\lambda_k) \in \mathbb{R}^{k\times k}, \qquad V_k = [v_1 \mid \cdots \mid v_k] \in \mathbb{R}^{n\times k}.$$

Why truncate at k? Retaining the top k eigenpairs gives the best rank-k approximation to B in the Frobenius norm (Eckart–Young theorem, Appendix D), and hence the embedding that minimises the reconstruction error in the inner-product sense.

Step 4: Recover the Coordinates

We need $Z_k \in \mathbb{R}^{n\times k}$ such that $Z_k Z_k^\top \approx B$. From the truncated decomposition:

$$B \approx V_k \Lambda_k V_k^\top = \big(V_k \Lambda_k^{1/2}\big)\big(V_k \Lambda_k^{1/2}\big)^\top.$$

Hence we set

$$\boxed{Z_k = V_k\, \Lambda_k^{1/2},} \tag{7}$$

where $\Lambda_k^{1/2} = \operatorname{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_k})$.
The i-th row of Z_k gives the k-dimensional embedding of object i:

$$z_i = \Lambda_k^{1/2} V_k^\top e_i \in \mathbb{R}^k,$$

where e_i is the i-th standard basis vector.

Conclusion

Starting from pairwise squared dissimilarities $D_{ij} = d_{ij}^2$, cMDS recovers low-dimensional coordinates in three steps:

1. **Double-centre** D to obtain the Gram matrix of inner products:

$$B = -\tfrac{1}{2}\, HDH.$$

2. **Eigendecompose** B; retain the k largest eigenpairs (λ_i, v_i).

3. **Form coordinates** $Z_k = V_k \Lambda_k^{1/2}$, whose rows are the desired k-dimensional embeddings.

cMDS embeds objects in $\mathbb{R}^k$ by matching pairwise Euclidean distances to the given dissimilarities via inner-product factorisation.

A Double Centring in Matrix Form

Let $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$. For any matrix A, the (i, j) entry of HAH is:

$$(HAH)_{ij} = A_{ij} - \frac{1}{n}\sum_{j'} A_{ij'} - \frac{1}{n}\sum_{i'} A_{i'j} + \frac{1}{n^2}\sum_{i',j'} A_{i'j'}.$$

Applying this with $A = D$ and multiplying by $-\frac{1}{2}$ recovers exactly formula (5), confirming $B = -\frac{1}{2}HDH$.

B Symmetric Positive Semidefinite (PSD) Matrices

A matrix $M \in \mathbb{R}^{n\times n}$ is **symmetric positive semidefinite** if:

- $M = M^\top$ (symmetry), and
- $x^\top M x \geq 0$ for all $x \in \mathbb{R}^n$.

Equivalently, all eigenvalues of M are non-negative.
$B = ZZ^\top$ is always PSD: $x^\top B x = x^\top Z Z^\top x = \|Z^\top x\|^2 \geq 0$.
If the given dissimilarities are *not* Euclidean, some eigenvalues of B may be (slightly) negative due to noise; in practice these are set to zero.

C Spectral Theorem (Eigendecomposition of Symmetric Matrices)

Theorem. Every real symmetric matrix $M \in \mathbb{R}^{n\times n}$ can be written as

$$M = V\Lambda V^\top,$$

where $V \in \mathbb{R}^{n\times n}$ is orthogonal ($V^\top V = I$) and $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ contains the real eigenvalues of M.
For a PSD matrix, all $\lambda_i \geq 0$, so $\Lambda^{1/2}$ is well-defined.

D Eckart–Young Theorem (Best Low-Rank Approximation)

Theorem. Let $M = V\Lambda V^\top$ be the eigendecomposition of a symmetric PSD matrix, with eigenvalues ordered $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$. Among all rank-$k$ matrices $\hat{M}$, the one minimising $\|M - \hat{M}\|_F^2$ is

$$\hat{M} = V_k \Lambda_k V_k^\top,$$

where V_k and Λ_k retain only the top k eigenpairs.
In cMDS this justifies choosing the k largest eigenvalues: the resulting embedding minimises the total squared error between B and $Z_k Z_k^\top$.